Quiz 3

Graded

Student

Myles Sartor

Total Points

16 / 18 pts

Question 1

Problem 1

4 / 4 pts

✔ **– 0 pts** Correct

– 1 pt one of the parameters is incorrect

– 3 pts major errors in problem

– 2 pts errors in expressing probability

– 1 pt error in taking the complement

– 4 pts no answer

Question 2

(no title)

6 / 8 pts

– 0 pts Correct

– 1 pt P(Tail) formula

– 1 pt P(Tail) answer

– 1 pt (a) E(x) = np for binomial distribution

– 1 pt (a) E(X) answer

– 1 pt (b) V(X) = np(1-p) for binomial distribtion

– 1 pt (b) V(X) answer

✔ **– 1 pt** (c) V(Y) = 16V(X)

✔ **– 1 pt** (c) V(Y) answer

– 8 pts Incorrect

Question 3

(no title)

6 / 6 pts

✔ **− 0 pts** Correct

− 1 pt (a) binomial distribution

− 0.5 pts (a) parameters 10 for n

− 0.5 pts (a) 0.2 for p

− 2 pts (b) $P(X=0)=\binom{10}{0}(0.2)^0(0.8)^{10}$

− 0.5 pts (c) $\sum_{k=0}^{3}\binom{10}{k}(0.2)^k(0.8)^{10-k}$: forgot to include 0 sales

− 0.5 pts (c) $\sum_{k=0}^{3}\binom{10}{k}(0.2)^k(0.8)^{10-k}$: forgot to include 3 sales

− 2 pts (c) $\sum_{k=0}^{3}\binom{10}{k}(0.2)^k(0.8)^{10-k}$

− 1.5 pts (c) $\sum_{k=0}^{3}\binom{10}{k}(0.2)^k(0.8)^{10-k}$

− 1 pt (b) $P(X=0)=\binom{10}{0}(0.2)^0(0.8)^{10}$

− 1 pt probability is 0.2 not 20

− 1 pt (c) incorrect pmf

− 1 pt (c) should not take the compliment

− 1 pt (c) incorrect indices

Quiz 3

Name: Myles Sartor Student ID: 119c17708

Problem 1. A random variable X has a binomial distribution with mean 2 and variance 1.6. Find probability that X is at least 2. → complement is X is at most 2

$P(X \geq 2)$?

or

$1 - P(X \leq 2)$?

$E[X] = \mu = np = 2$

$Var(X) = np(1-p) = 1.6$

$2(0.8)$

$\frac{10}{2} \cdot \frac{2}{1} = 10$

$2(1-p) = 1.6$

$1-p = 0.8$

$\frac{10}{1} \cdot \frac{2}{10} = 2$

$p = 0.2$

$n \cdot 0.2 = 2$

$n = 10$

$$P(X \geq 2) = \sum_{X=2}^{\infty} \binom{10}{X} (0.2)^X (0.8)^{10-X}$$

or

Complement:

$$P(X \geq 2) = 1 - \sum_{X=0}^{2} \binom{10}{X} (0.2)^X (0.8)^{10-X}$$

↳ Allows approximation of real value

Quiz 3

Name: Myles Sartor Student ID: 119017708

0.9

✱ **Problem 2.** Suppose we have two coins. Coin C_1 comes up heads with probability 0.1 and coin C_2 comes up heads with probability 0.6. Probability of choosing coin C_1 is $\frac{1}{3}$ and probability of choosing coin C_2 is $\frac{2}{3}$.

0.4

$1-0.1=0.9=P(\text{Tail}|C_1)$

$1-0.6=0.4=P(\text{Tail}|C_2)$

We repeat the following process 3 times:

- Choose a coin.
- Flip that coin once.

$n=3$

$P(\text{Tail}) = P(\text{Tail}|C_1)P(C_1) + P(\text{Tail}|C_2)P(C_2)$

$0.9 \cdot \frac{1}{3} + 0.4 \cdot \frac{2}{3}$

Suppose X is the number of Tails after 3 flips.

a. What is E[X]?

b. What is Var(X)?

$\frac{9}{10} \cdot \frac{1}{3} + \frac{4}{10} \cdot \frac{2}{3} = \frac{9}{30} + \frac{8}{30} = \frac{17}{30}$

c. Based on the number of tails we get, we earn $Y = 4x - 1$. What's Var(Y)? $P = \frac{17}{30}$

a) $E(X) = np \to 3\left(\frac{17}{30}\right) = \frac{3}{1} \cdot \frac{17}{30} = \frac{51}{30} = \frac{17}{10}$

$\frac{30}{30} - \frac{17}{30} = \frac{13}{30}$

b) $Var(X) = np(1-p) \to \frac{17}{10}\left(1 - \frac{17}{30}\right)$

$\frac{17}{10} \cdot \frac{13}{30} = \frac{221}{300}$

$\nearrow 4 \cdot Var(X) - 1$

✱ c) $Var[Y] = var(4x-1) = \sum_{x=}^{3} 4x-1$

$E[x^2] - (E[x])^2$

$\left(\frac{17}{10}\right)^2$

$\frac{4}{1} \cdot \frac{221}{300} - 1$

$\frac{4}{1} \cdot \frac{221}{300} - \frac{300}{300}$

Quiz 3

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$p=0.20$

Problem 3. Sarah works as a sales representative selling house insurance over the phone. Only 20% of all calls result in a successful sale. Assume that the outcome of each call is independent of the others.

If Sarah makes 10 calls, and X is the number of sales that she makes

(a) what is distribution of X? write down your full answer with specifying the parameters as well.

(b) what is the probability that she does not make any sale?

(c) what is the probability that she makes at most 3 sales?

a) $P_X(x) = \binom{10}{x}(0.20)^x(0.80)^{10-x}$ $\quad X \sim Bin(n, p)$

b) $P(X=0) = \binom{10}{0}(0.20)^0(0.80)^{10}$

$\rightarrow P(X=0)+P(X=1)+P(X=2)+P(X=3)$

c) $P(X\leq 3) = \sum_{X=0}^{3} \binom{10}{X}(0.20)^X(0.80)^{10-X}$

$\hookrightarrow$ Complement is making more than 3 sales (at least 3)

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If $X \sim Bin(n,p)$, then $X = 0, 1, 2, \cdots, n$ and

$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$, $E[X] = np$, $Var(X) = np(1-p)$.

If $X \sim Bern(p)$, then $X = 0, 1$ and

$P(X = k) = p^k (1-p)^{1-k}$, $E[X] = p$, $Var(X) = p(1-p)$.